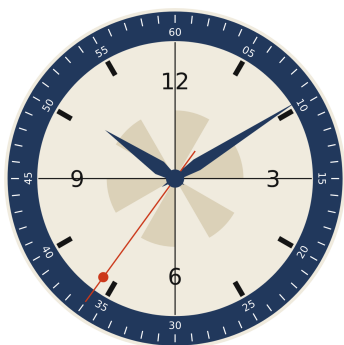


DIAL TOOL

A watch-dial design tool for DrawBot on macOS. Guidebook · beta 2.3 · 4 Oct 2026



A sector dial



A 270° scale



Imported marker and hand shapes

WHAT IT DOES

- **Rings.** The dial is a list of rings – ticks, markers, numerals, bands of colour. As many as you like: reorder, hide, duplicate, rename.
- **Positions.** Any count, any start, any span. Every n-th. Scales and arcs.
- **Give way.** A ring can skip the positions of the rings above it, or knock a gap round them.
- **Type.** Any installed font, per numerals ring: styles, variable axes, OpenType features, language, tracking.
- **Hands.** Ten shapes or your own drawing. Lengths in mm, lume openings, counterweights; sweeping or beating seconds.
- **Files in.** SVG, PDF and AI shapes for markers and hands.
- **Files out.** PDF, SVG, PNG at any dpi, MP4, GIF – true size in mm. Outlines, or live text.
- **Production.** A one-colour print mask, mirrored for toner transfer, with the date aperture as a cut line.

IN ONE MINUTE

1. Install DrawBot (free, macOS).
2. Open `dial.py` in DrawBot.
3. Press `⌘R`. Change anything; the preview follows.

Everything is in millimetres. Angles are clock angles: 0° is 12 o'clock, growing clockwise.

TYPING NUMBERS

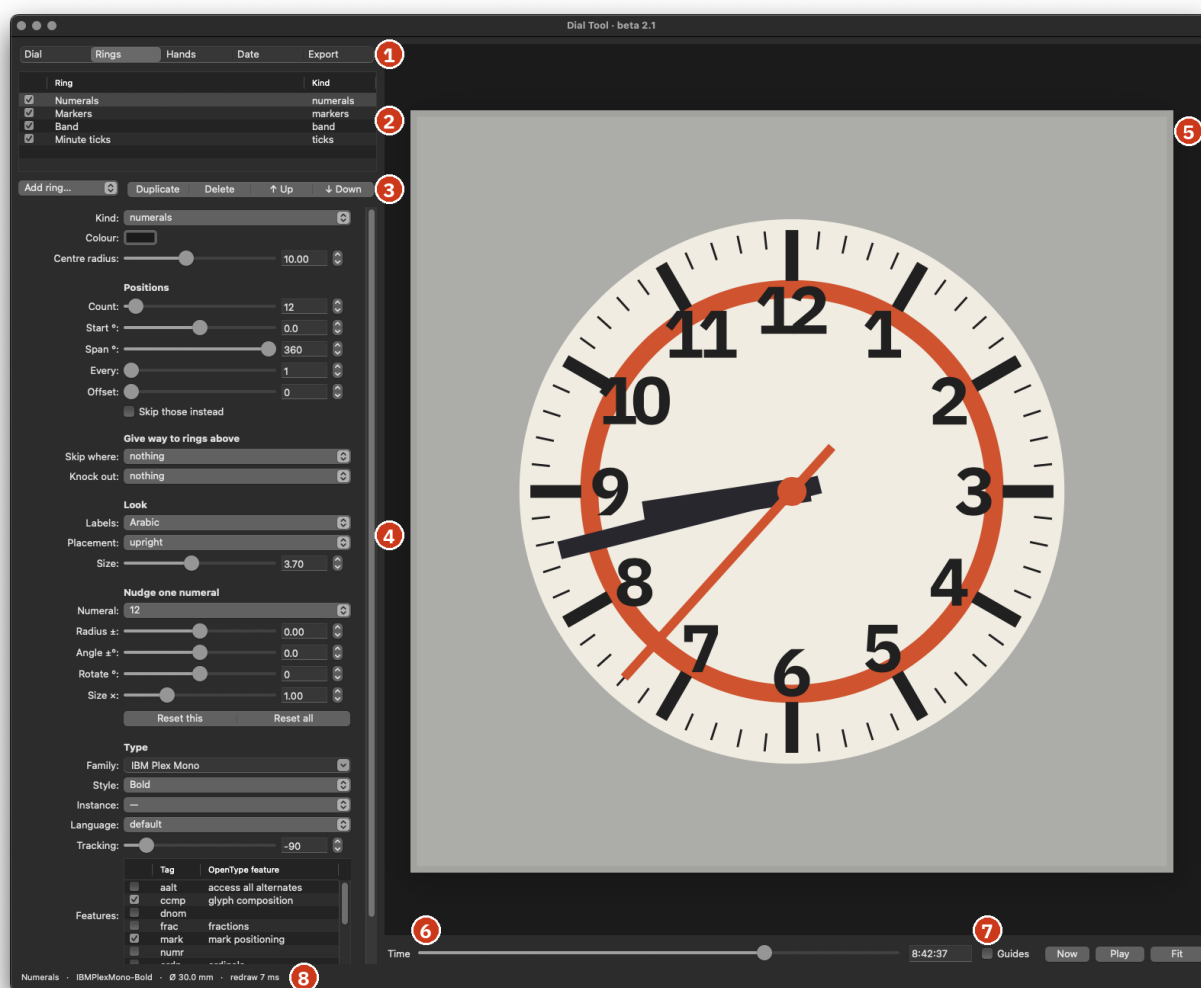
Every number has a slider, a field and a stepper. In the field: `1.2` sets · `+0.5` adds · `x0.5` multiplies · `/2` divides. Typing past a slider's end extends the slider.

INSTALL AND FIRST RUN

INSTALL

1. **DrawBot 3.132 or newer** – free, macOS only. drawbot.com → Download.
 2. **dial.py** – github.com/duztman/dial-tool → [dial.py](#) → download the raw file. Branch **main** is released; **dev** is the newest work.
 3. Open **dial.py** in DrawBot and press **⌘R**. The Dial Tool window opens.
 4. Optional, for speed: DrawBot → Python → Install Python Packages → **skia-pathops** → Go.
- Pressing **⌘R** again replaces the window and keeps your settings. Quitting DrawBot keeps them too: the last session reopens next time.

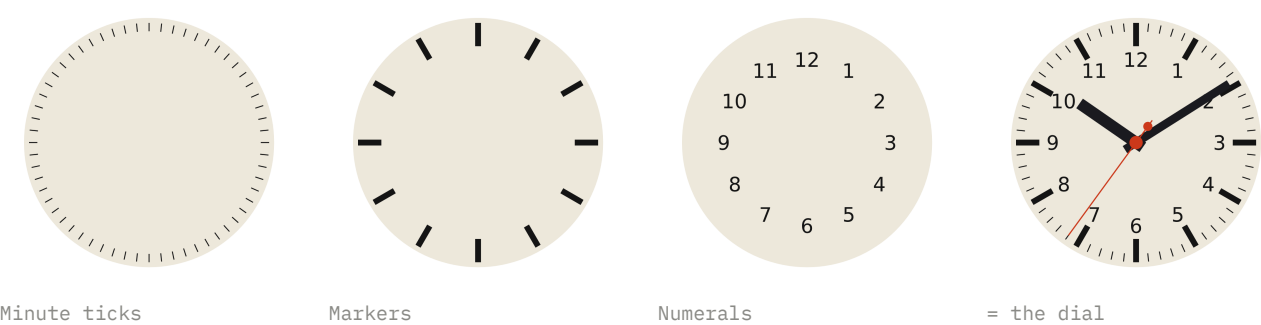
THE WINDOW



- | | |
|--|---|
| 1 Sections – Dial · Rings · Hands · Date · Export | 5 Preview – pinch or ⌘ -scroll zooms, two fingers pan, Fit resets |
| 2 Ring list – show · name (double-click to rename) · kind | 6 Time – drag, or type 10:09:36 ; Now; Play |
| 3 Add ring... · Duplicate · Delete · Up · Down | 7 Guides – construction circles, never exported |
| 4 The selected ring's settings – one scrolling list | 8 Status – ring · font · diameter · build and frame time |

RINGS

The dial is a list of rings. Each ring repeats one thing around the centre, at its positions. The list order is the stacking order: the top ring is drawn last, in front.



KINDS

ticks	the minute track, dots, a crosshair
markers	bars, wedges, dots or your own shape; a special 12 o'clock (double, triangle, none)
numerals	Arabic, Roman, Eastern Arabic, Chinese, words, numbers from a start, or your own list
band	a ring of colour; sectors; pies to the centre

Change a ring's kind at any time – its settings stay. **Radius** is the outer edge (numerals: the text's centre).

POSITIONS

- **Count** – how many. **Start °** – where the first sits. **Span °** – 360 for all round; less makes an arc with both ends marked.
- **Every** n-th, from **Offset**. **Skip those instead** turns it round. Cardinals: every 3. The other hours: every 3, skip those instead.

GIVE WAY TO RINGS ABOVE

- **Skip where** – leave out positions where markers, markers & numerals, or any marks above already sit.
- **Knock out** – cut a gap of **Clearance** mm round markers, numerals, or everything above.

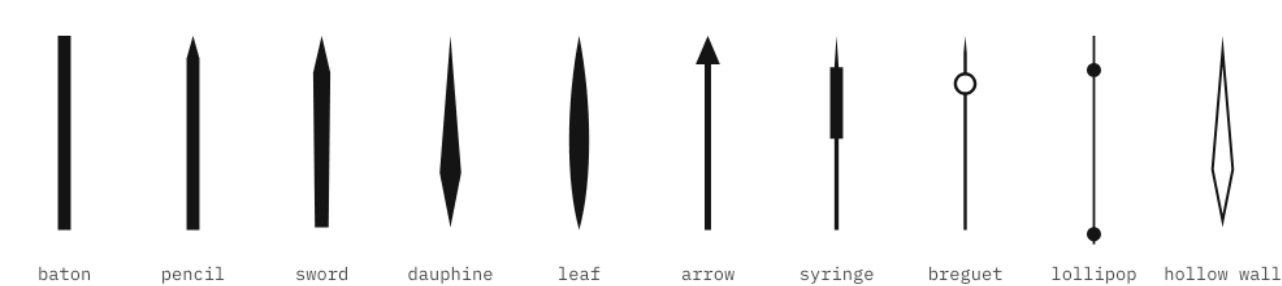
RECIPES

- **Railroad track** – ticks (count 60), and two thin bands: one at the ticks' radius, one at radius – tick length + band width.
- **Fifths of a second** – ticks count 60 above ticks count 300; the 300 skip where: all marks above.
- **Minute numerals 05 ... 60** – numerals, labels Numbers, first 5, step 5, digits 2, start 30°.
- **Sector dial** – a band with count 12, every 2, band width = radius: alternate pies.
- **A 270° scale** – numerals count 10, start -135°, span 270°, Numbers from 0, step 10.
- **A gap round numerals on a band** – band below the numerals; knock out: markers & numerals above.

HANDS, TYPE, FILES

HANDS

Choose Hour, Minute or Seconds, then a shape. Lengths are millimetres from the pivot.



Each shape adds its own controls: tip length, head width, ring or disc and where it sits. **Hollow wall** cuts a lume opening; a **counterweight** fits any tail. Seconds move as a sweep, quartz (1/s), or 6 or 8 beats per second.

TYPE

Each numerals ring has its own type: family → style → named instance; variable axes as sliders; OpenType features with the designer's names; language (Turkish İ ı); tracking in 1/1000 em, like Illustrator. The date window uses the top numerals ring's type.

FILES IN – SVG • PDF • AI

- Draw pointing up (12 o'clock). The artboard's centre is the anchor: a hand's pivot, a marker's centre.
- Black is ink. White shapes cut holes. Strokes become outlines.
- Outline text first; images and gradients are skipped, and the Export log says so.
- Save AI files with "Create PDF Compatible File". For hands: **Fit file to length**, or **Size x**; **Width x** stretches.

FILES OUT – THE EXPORT SECTION

PDF	vector, true size in mm – Illustrator, print, the press
SVG	vector, true size – Illustrator, laser, the web
PNG	raster at the dpi you set – previews, film for toner transfer
MP4 • GIF	the hands moving from the set time – Clip seconds, Frames per s

- **Production** – one-colour mask: no plate, no hands, the aperture as a cut line.
- **Mirror** – flipped, for toner transfer.
- **Live text** – numerals stay text in PDF and SVG. Off by default: outlines need no fonts and print exactly as seen.
- **Settings** – Save... and Load... a **.json** of everything; Defaults starts over.

NOTES • FAQ • HELP

FAQ

- **Where are my settings?** In `~/Library/Application Support/DialTool/last.json`, saved as you work. Export → Save... keeps a named copy.
- **I changed the diameter and nothing moved.** Radii and hand lengths are millimetres from the centre, as on a real movement. A "scale everything" action is on the list.
- **Outlines or live text?** Outlines, unless you need to edit text later. With live text, Illustrator may still outline numerals that use ligatures or alternate glyphs, and on-path numerals become one text per letter.
- **Which fonts can I use?** Any font installed on the Mac. There is no fonts folder.
- **A ring disappeared.** Check its show checkbox, its Every and Offset, and whether it skips positions a ring above takes.
- **A numeral is missing at the date.** Markers and numerals leave the date's position empty, and all print keeps the Print clearance from the hole.
- **How do I start over?** Export → Settings → Defaults.
- **Canvas or PDF?** The switch under the preview. Canvas is fast and sharp; PDF is DrawBot's own viewer, the reference. They should look the same.
- **Do my beta 1.0 files still open?** Yes – they open as rings, unchanged.

WHEN SOMETHING GOES WRONG

1. Open the Export section and read the log. At launch it says `preview: canvas` and `all controls connected`.
2. Send a screenshot and the log.
3. If the window doesn't build at all, run `tools/dial_probe.py` in DrawBot and send its report.

Draw at true size. A 30 mm dial at 600 dpi is 709 pixels across; thin lines need room: check the thinnest one at 100 % before you etch or print.

WHERE THINGS LIVE

- **Code and docs** – github.com/duztman/dial-tool (`main` released • `dev` in progress).
- **Claude Project "Dial Tool"** – reads the same repository.
- **This guide** – `docs/guide/guide.md`, built by `tools/make_guide.py` into `docs/Dial-Tool-guide.pdf`, updated with each release.
- **Words** – ring, kind, positions, give way, band, nudge: `docs/glossary.md`.